NTSE STAGE – I (2019 – 20) HARYANA STATE
(For Class – X)
MATHEMATICAL APTITUDE TEST (MAT)

QUESTIONS (SET – C)

Directions (Questions 1 – 5): Each letter of alphabet from A to Z has been give a value from 1 to 26 serially. Solve the questions on the basis of value of words.

1. BUSH = 50 CAMP = 33 then LIKE = ?
   (1) 40 (2) 41
   (3) 32 (4) 37

2. Which word has the maximum value?
   (1) BURN (2) CURT
   (3) DUCK (4) BUOY

3. Which words have the equivalent values?
   (1) KING : CAST (2) BURY : SURE
   (3) RICH : BOAT (4) BLUE : CANT

4. Which equation is correct?
   (1) $X + Y = 50$ (2) $Z - T = 6$
   (3) $B \times V = 41$ (4) $R \div I = 5$

5. Which word is equivalent to 106?
   (1) MONKEY (2) DOG JACKY
   (3) HAI HAPPY (4) SO LUCKY

6. What will come at the place of?

T	**E**	**Y**
O	**K**	**Z**
R	**G**	**?**

   (1) W (2) X
   (3) Y (4) Z

7. How many times in 24 hours the hands (hour and minute) of a clock will be at right angles?
   (1) 24 (2) 30
   (3) 72 (4) 48

8. If a train runs at a speed of 92.7 km/hr, then the distance covered in metres in 17 minutes will be:
   (1) 26265 (2) 26700
   (3) 30002 (4) 29365

9. Here are some words translated from an artificial language –
   mie pie is blue light
   mie tie is blue berry
   aie tie is rasp berry
   which words could possibly mean ‘light fly”
   (1) pie zie (2) pie mie

(3) aie zie (4) aie mie

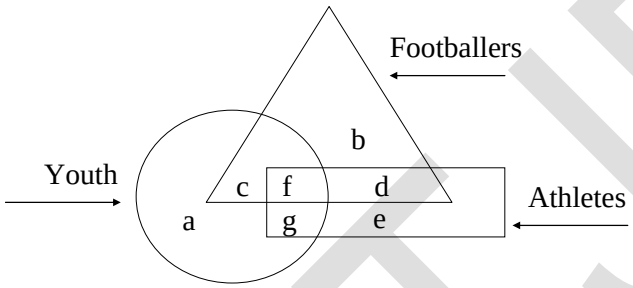
10. If in a certain code, STUDENT is written as RSTEDMS, then how would TEACHER be written in the same code?
(1) SZZDGEQ (2) SZDDGEQ
(3) SDZDGDQ (4) SDZCGDQ

11. If CHAIR is coded as FKDLU then RAID is coded as:
(1) ULGD (2) ULKG
(3) ULDG (4) UDLG

12. In a certain code HNDT has been coded as 3694. How will you code THD in the same code?
(1) 604 (2) 428
(3) 439 (4) 349

13. If the word PENCIL is coded as LICNEP then how would the word INKOPT be coded?
(1) TOPINK (2) JOLQPU
(3) HMKOPS (4) TOPKNI

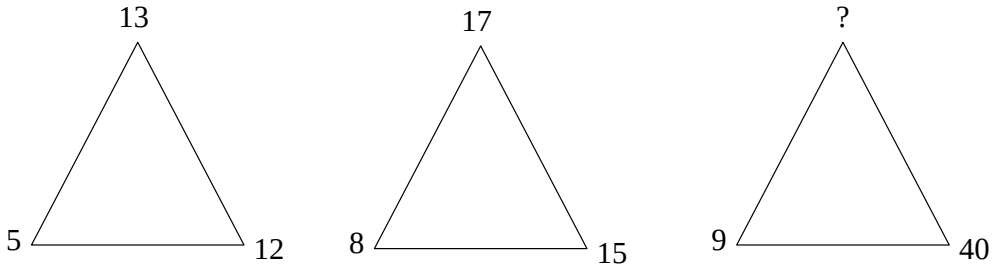
14. In the figure, the circle represents youth, the triangle represents footballers and the rectangle represents athletes – which letter represents athletes among youth who are not footballers?



(1) c (2) g
(3) d (4) f

15. Statement: All clocks are Alarms. No Clocks are cuckoos. All cuckoos are Alarms. Some cuckoos are Birds.
Conclusion:
(I): Some Alarms are Birds.
(II): No clock is a Birds.
(III): All Birds are Alarms.
(1) Only conclusion I follows (2) Only conclusion II follows
(3) Only conclusion III follows (4) Both conclusion II and III follows

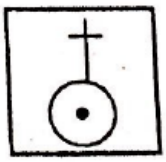
16. Find the number in the position of '?'



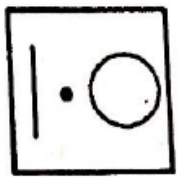
(1) 42 (2) 40
(3) 41 (4) 45

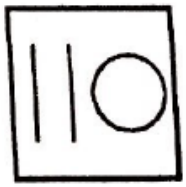
Directions (Questions 17 – 20): Find out which of the alternatives will exactly make up the key figure (X)?

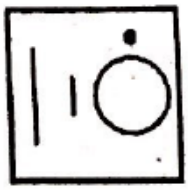
17.

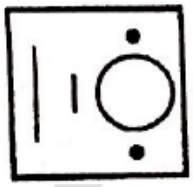


(X)

(1) 

(2) 

(3) 

(4) 

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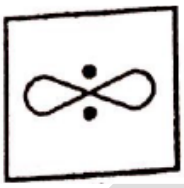
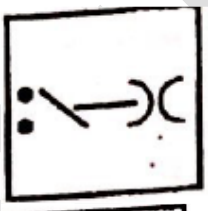


(X)

(1) 

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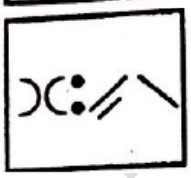
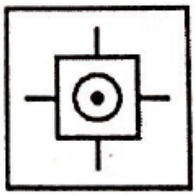
(3) 

(4)

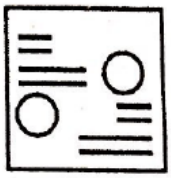
19.

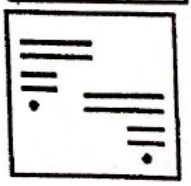


(X)

(1)

(2)

(3) 

(4) 

20.

(X)

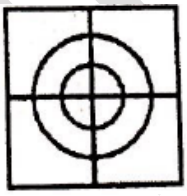
(1)

(2)

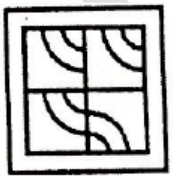
(3)

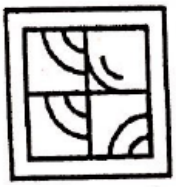
(4)

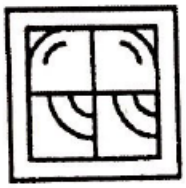
Directions (Questions 21 – 23): Which figure is the rearrangement of the parts of the given figure?

21. 

(X)

(1) 

(2) 

(3) 

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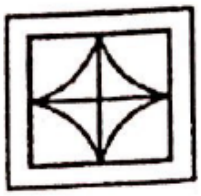
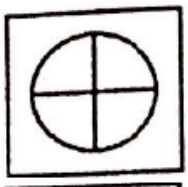


(X)

(1)



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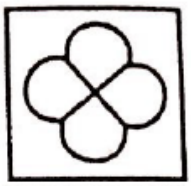

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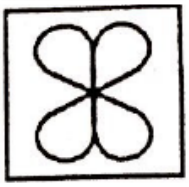

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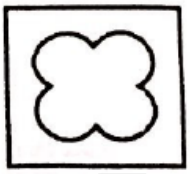

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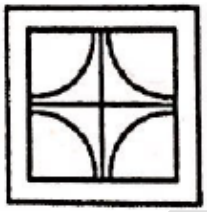
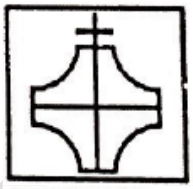


(X)

(1)



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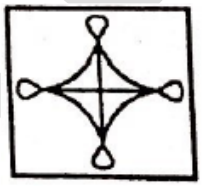

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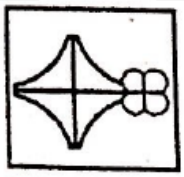

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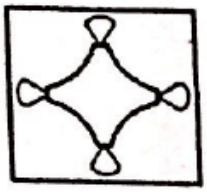
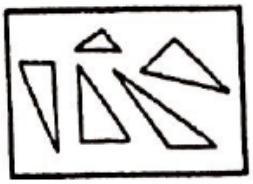


FIITJEE

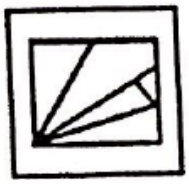
Directions (Questions 24 – 28): Find out which of the figure (1, 2, 3 and 4) can be formed form the piece given in the figure (X).

24.



(X)

(1)



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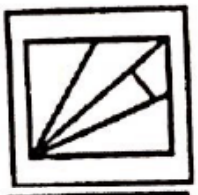
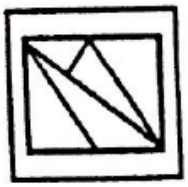


(3)



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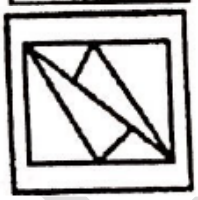

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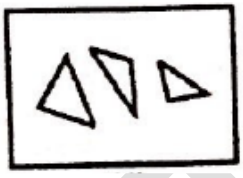
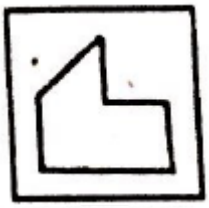


(X)

(1)



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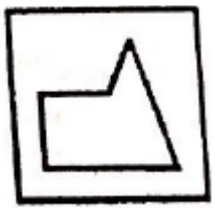

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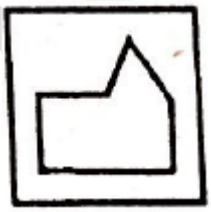

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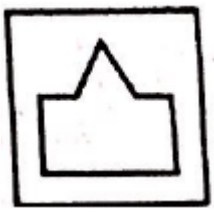

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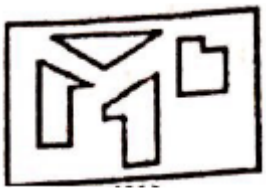
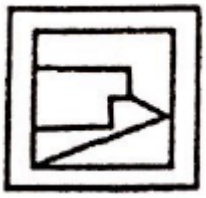


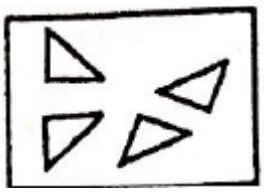
(X)

(1)

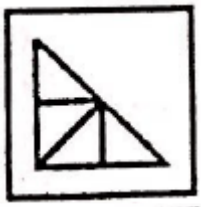
(2)

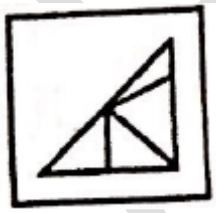
(3)

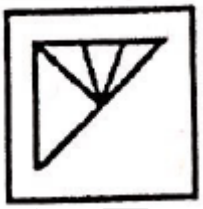
(4) 

27. 

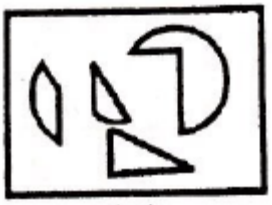
(X)

(1) 

(2) 

(3) 

(4)

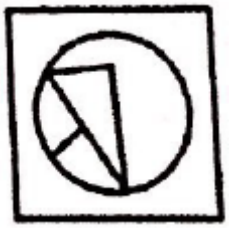
28. 

(X)

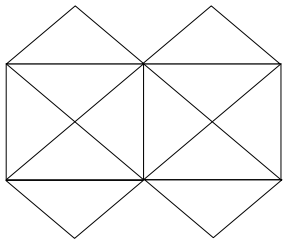
(1)

(2)

(3)

(4) 

Directions (Questions 29 – 31): Study the following figure and answer the questions.

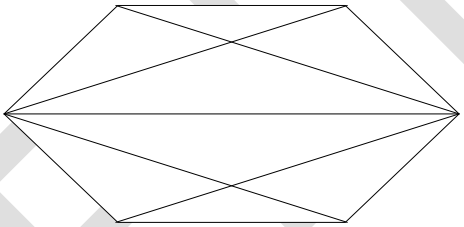


29. What is the minimum number of straight lines that is needed to construct the figure?
(1) 11 (2) 13
(3) 15 (4) 21

30. Count the number of triangle in the figure?
(1) 22 (2) 16
(3) 20 (4) 24

31. How many squares does the figure contain?
(1) 5 (2) 6
(3) 7 (4) 8

Directions (Questions 32 – 33): Analyze the following figure and answer the questions.



32. Find the number of quadrilaterals.
(1) 6 (2) 7
(3) 9 (4) 10

33. Find the number of Pentagons.
(1) 2 (2) 3
(3) 4 (4) 6

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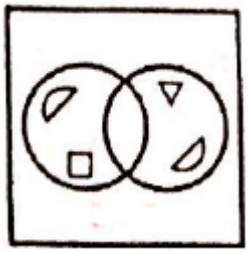
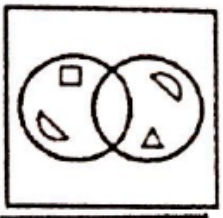
Directions (Questions 34 – 38): Choose the mirror – image of the figure (X) from amongst the four alternatives 1, 2, 3 and 4 given along with it.

34.

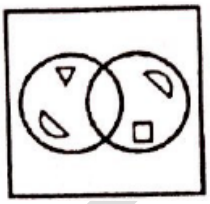


(X)

(1)



(2)



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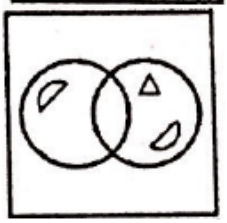

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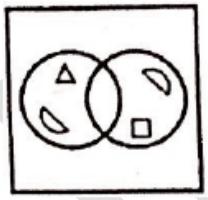

35.

(X)

(1)

(2)

(3)

(4)

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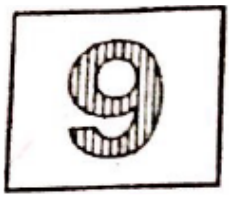
36.

(X)

(1)



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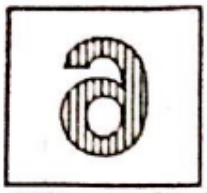

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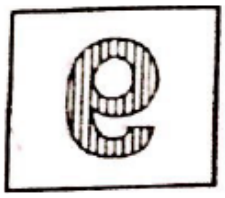
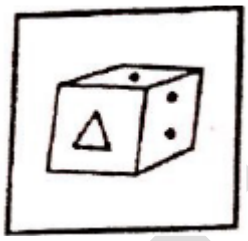


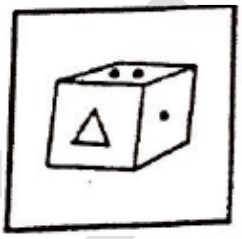
(4)

37.



(X)

(1)



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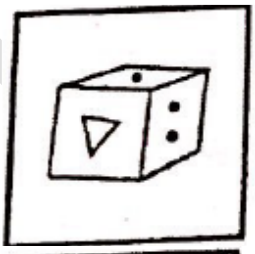

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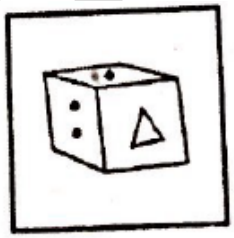

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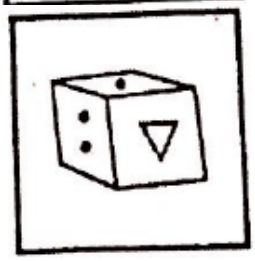
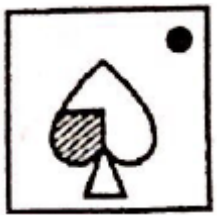
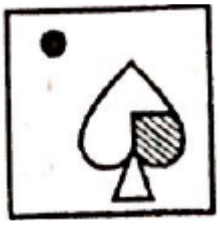
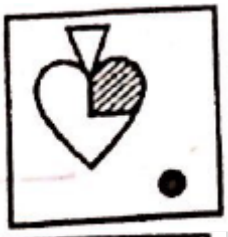


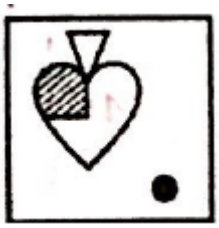
38.

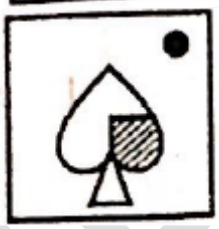


(X)

(1) 

(2) 

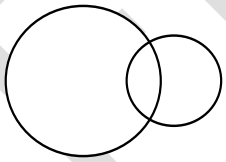
(3) 

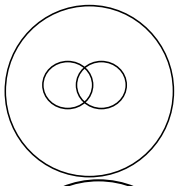
(4) 

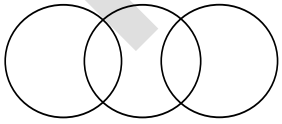
39. Asha's house faces south direction. She runs ahead 10m, turns left and runs 5m. Then she turns left again and turn 15m and then again turns left and runs 10m. Finally she turns right and runs 5m to reach her friend's house. What direction is Asha's friend's house facing?
(1) South (2) North
(3) East (4) West

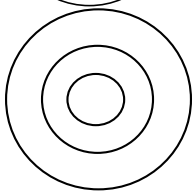
40. Mother was asked how many gifts she had in the bag. She replied that there were all dolls except six, all cars but six and all books but six. How many gifts had she in all?
(1) 36 (2) 27
(3) 18 (4) 9

41. Which of the following diagram indicate the relation between women, mothers and parents?

(1) 

(2) 

(3) 

(4) 

42. In a dairy, there are 60 cows and buffalos. The number of cows is twice that of buffalos. Buffalo X ranked seventeenth in terms of milk delivered. If there are 9 cows ahead of Buffalo X, how many buffalos are after in rank in terms of milk delivered?
(1) 10 (2) 11
(3) 12 (4) 13

43. Amongst five friends Lata, Alka, Rani, Asha and Sadhana. Lata is older than only three of her friends. Alka is younger to Asha and Lata. Rani is older than only Sadhana. Who amongst them is the eldest?

(1) Asha (2) Lata
(3) Alka (4) Sadhana

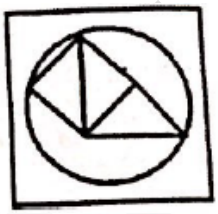
Directions (Questions 44 – 48): There is given a figure (X) followed by four figures 1, 2, 3 and 4 such that X is embedded in one of them. Trace out the correct alternatives.

44.

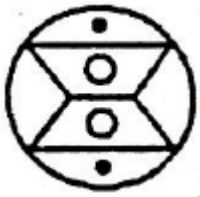
(X)

(1)

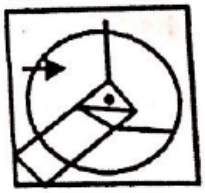
(2)



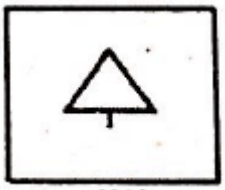
(3)



(4)

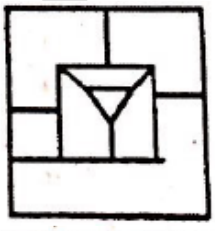


45.

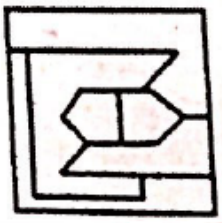


(X)

(1)



(2)



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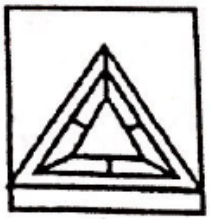

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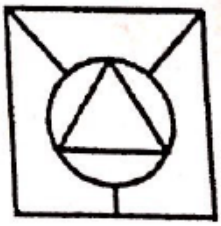
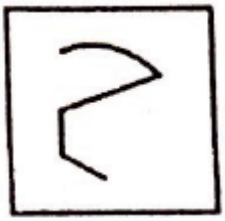
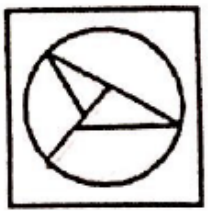


46.



(X)

(1)



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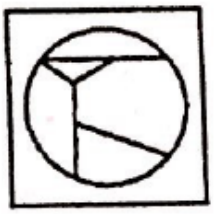

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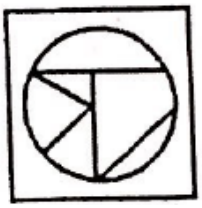

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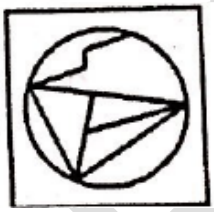
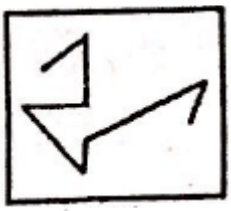
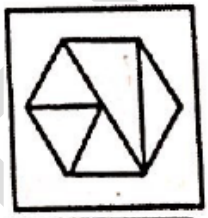


47.



(X)

(1)



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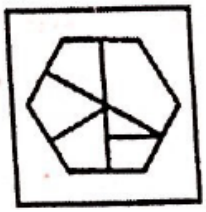

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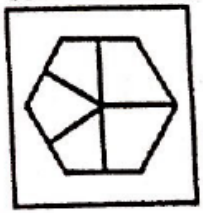

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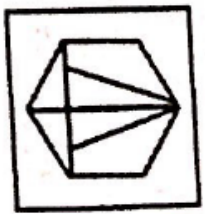

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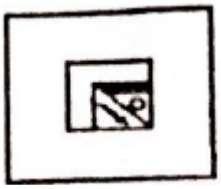
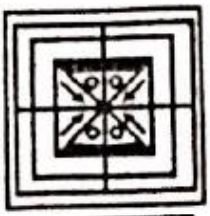
48.



(X)

(1)



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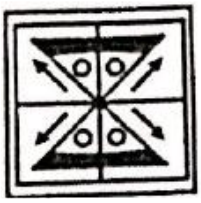
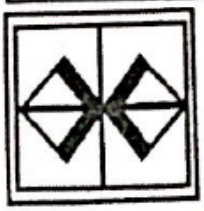
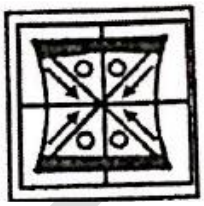


(3)

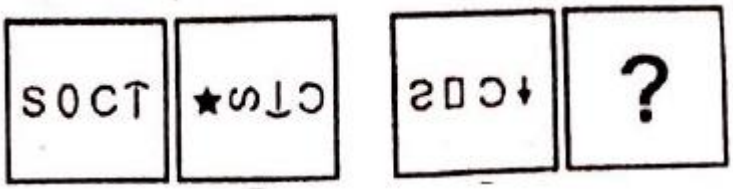


(4)



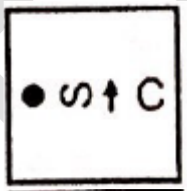
Directions (Questions 49 – 53): Figure A, B, C and D constitute the problem set while figure 1, 2, 3 and 4 constitute the answer set. There is a definite relationship between figures A and B. Establish a similar relationship between figures C and D by choosing a suitable figure (D) from the answer set.

49.



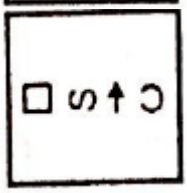
A B C D

(1)



(2)

(3)



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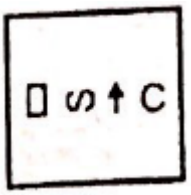
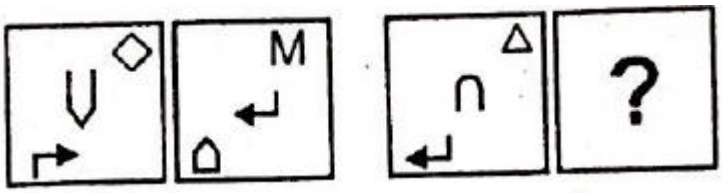
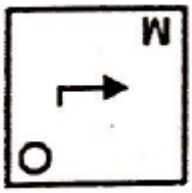
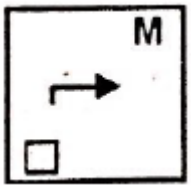


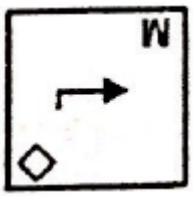
50.

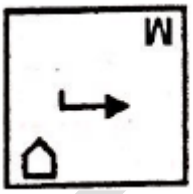


A B C D

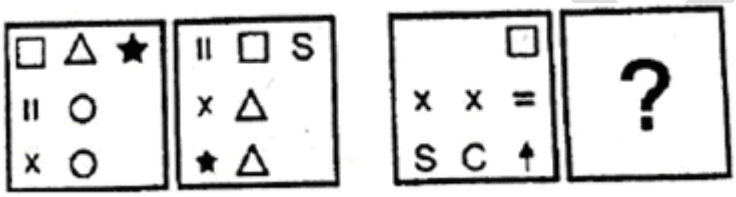
(1) 

(2) 

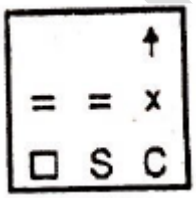
(3) 

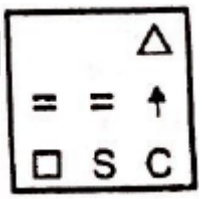
(4) 

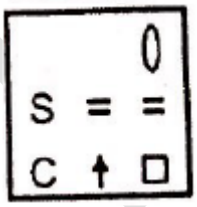
51.

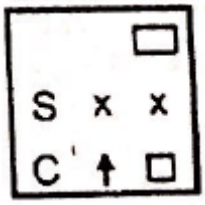


A B C D

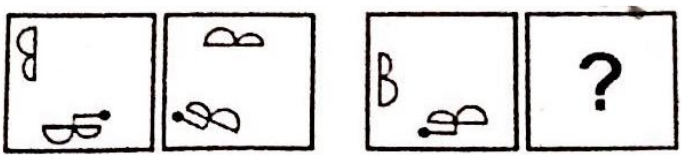
(1) 

(2) 

(3) 

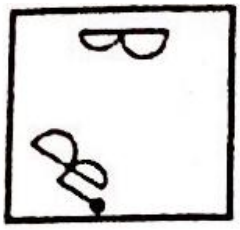
(4) 

52.



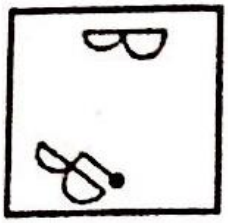
A B C D

(1)

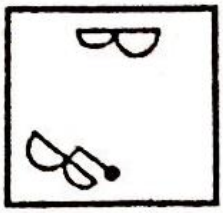


(2)

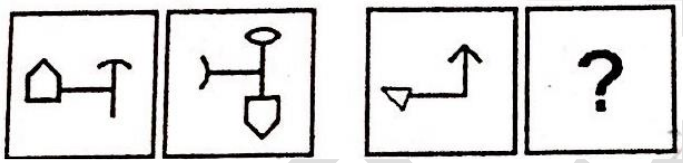
(3)



(4)

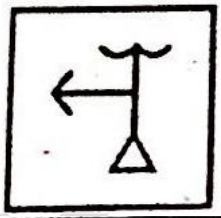


53.

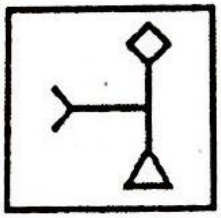


A B C D

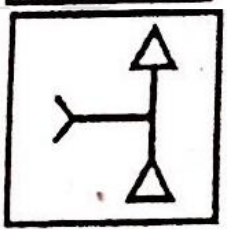
(1)



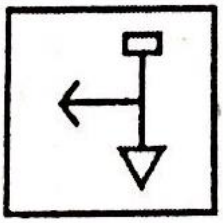
(2)



(3)

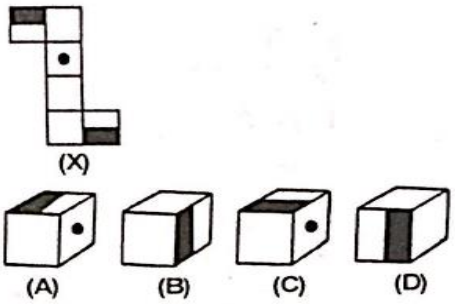


(4)



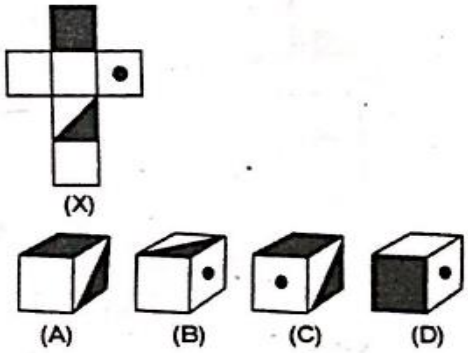
Direction (Questions 54 – 58): The figure (X) given in each problem, is folded to form a box. Choose from amongst the alternatives, the boxes that are similar to the box formed.

54.



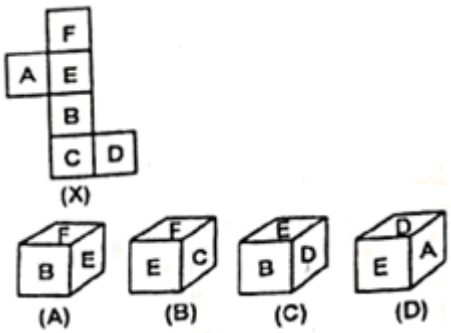
(1) A and B only
(2) B and C only
(3) B and D only
(4) A, B, C and D

55.



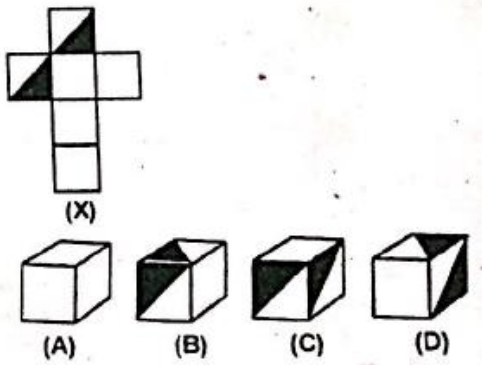
(1) A and C only
(2) B, C and D only
(3) B and D only
(3) C and D only

56.



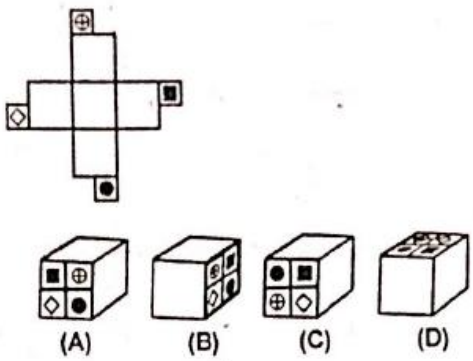
(1) A only
(2) C only
(3) A and C only
(4) A and B only

57.



(1) A and D only
(2) C and D only
(3) A and B only
(4) B and C only

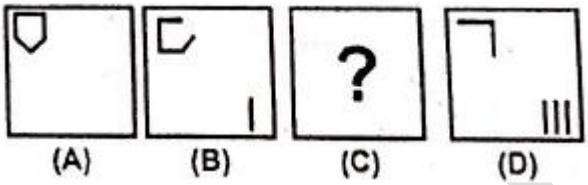
58.



(1) A and B only
(2) B only
(3) B and C only
(4) A, B and D only

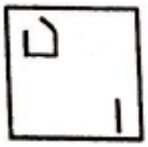
Direction (Questions 59 – 65): There are given a set of four figures (A, B, C and D) forming a certain series. However, the figure at C is missing. Choose this figure from the given alternatives.

59. **Problem Figures**



Answer Figures

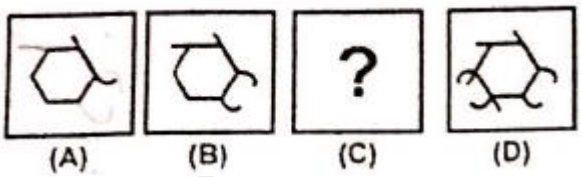
(1)

(2)


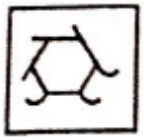
(3)

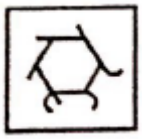
(4)

60. **Problem Figures**

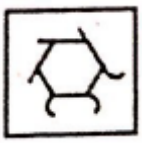


Answer Figures

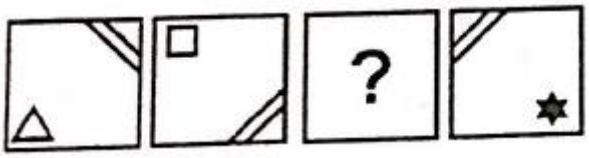
(1)


(2)


(3)

(4)


61. **Problem Figures**



Answer Figures

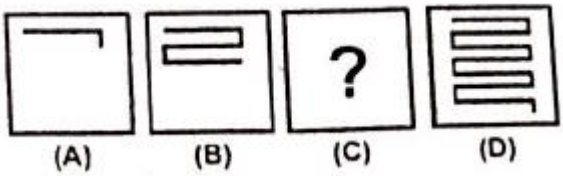
(1)

(2)

(3)

(4)

62. **Problem Figures**



Answer Figures

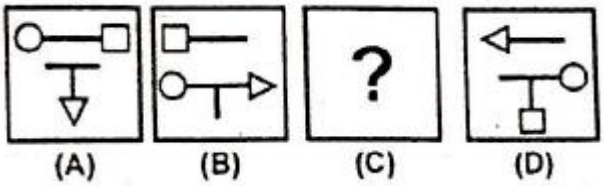
(1)

(2)

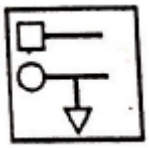
(3)

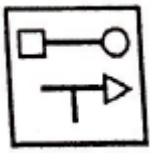
(4)

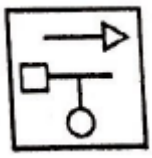
63. **Problem Figures**

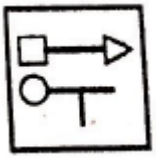


Answer Figures

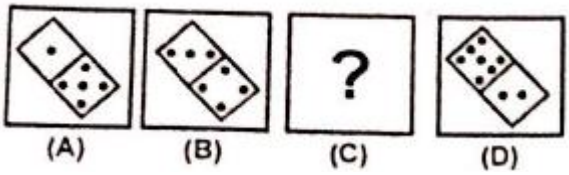
(1) 

(2) 

(3) 

(4) 

64. **Problem Figures**



Answer Figures

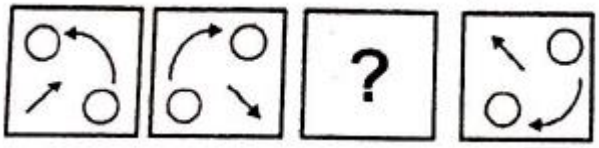
(1)

(2)

(3)

(4)

65. **Problem Figures**



Answer Figures

(1)

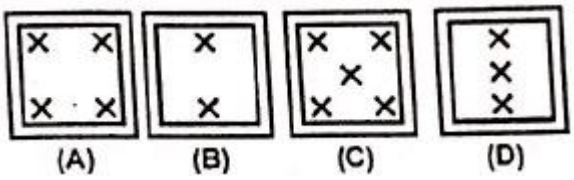
(2)

(3)

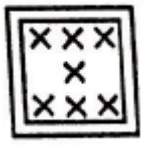
(4)

Direction (Questions 66 – 69): There are given four problem figures (A, B, C and D) and four Answer figures (1, 2, 3 and 4). Select a figure from amongst the answer figures which will continue the same series as given in the problem figures.

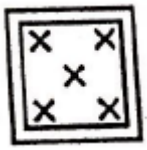
66. **Problem Figures**



Answer Figures

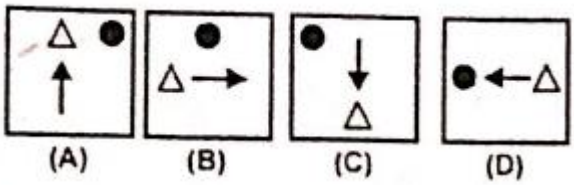
(1) 

(2)

(3) 

(4)

67. **Problem Figures**

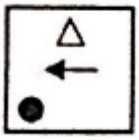


Answer Figures

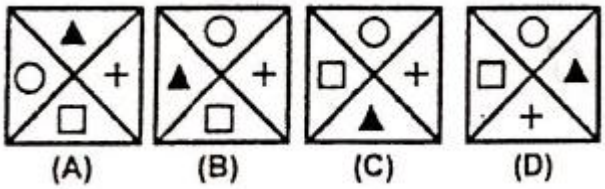
(1)

(2)

(3)

(4) 

68. **Problem Figures**



Answer Figures

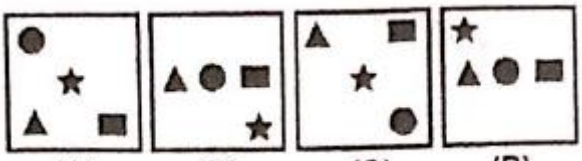
(1)

(2)

(3)

(4)

69. **Problem Figures**



(A) (B) (C) (D)

Answer Figures

(1)

(2)

(3)

(4)

70. Find the missing number in the series 2, 10, 26, 50, _____, 122.
(1) 81 (2) 82
(3) 80 (4) 84

Direction (Questions 71 – 74): In the questions, a series is given with one term missing. Choose the correct alternative from the given ones that will complete the series.

71. 6, 24, 60, 120, ?
(1) 180 (2) 210
(3) 240 (4) 360

72. 1, 9, 9, 81, 90, 810, 819, ?
(1) 7371 (2) 900
(3) 8100 (4) 1638

73. 2, 3, 6, 18, 108, ?
(1) 1944 (2) 1658
(3) 648 (4) 1008

74. 1, 2, 3, 4, 5, 7, 7, ?, ?
(1) 11, 13 (2) 10, 11
(3) 8, 9 (4) 9, 11

Direction (Questions 75 – 78): Find the wrong number in the following series.

75. 24576, 6144, 1536, 386, 96, 24
(1) 96 (2) 386
(3) 1536 (4) 6144

76. 3, 4, 10, 32, 136, 685
(1) 685 (2) 10
(3) 136 (4) 32

77. 3, 8, 13 24, 42, 70
(1) 13 (2) 24
(3) 42 (4) 70

FIITJEE

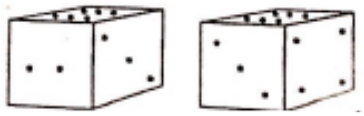
78. 6, 7, 9, 11, 15, 15, 28, 19, 36
(1) 15 (2) 28
(3) 19 (4) 70

79. Shitin remembers that his mother's birthday is after 17th April but before twenty first April, where as his father remembers that his wife's birthday is after 19th April but before 24th April. Which of the following days in April is definitely his mother's birthday?
(1) 19th April (2) 20th April
(3) 21st April (4) Both Statements are not sufficient

80. Day after tomorrow is my birthday. On the same day next week falls 'Holi'. Today is Monday. What will be the day after 'Holi'?
(1) Thursday (2) Friday
(3) Wednesday (4) Saturday

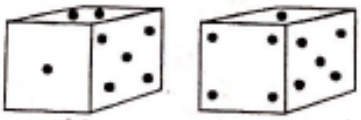
Direction (Questions 81 – 83): Two position of a block with 1 to 6 dots on its sides are shown below. Observe the dots on block.

81. If the block is resting on the side with three dots. What will be the number of dots on the side at the top?



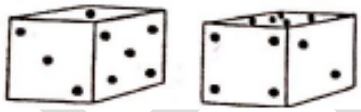
(1) 1 or 5 (2) 2
(3) 4 (4) 6

82. How many dots are contained on the face opposite to that containing four dots?



(1) 2 (2) 5
(3) 3 (4) 6

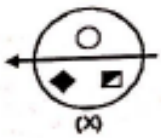
83. What is the number of dots on the face opposite 2 dots?



(1) 1 (2) 3
(3) 4 (4) 6

Direction (Questions 84 – 86): Choose the correct water image of the figure (x) from amongst the alternatives given along with its.

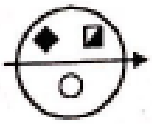
84.

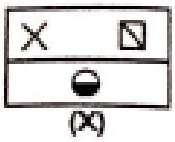


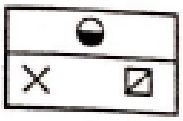
(1)

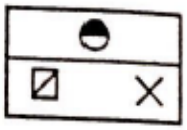
(2)

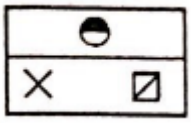
(3)

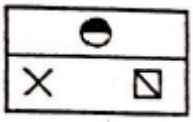
(4) 

85. 

(1) 

(2) 

(3) 

(4) 

86.

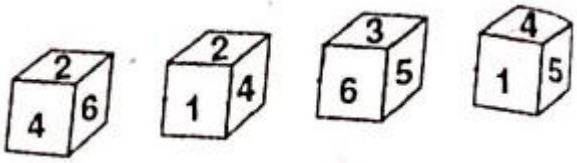
(1)

(2)

(3)

(4)

87. Which number is opposite to 3?



(1) 6 (2) 4
(3) 2 (4) 1

Direction (Questions 88 – 92): A cube painted red on two adjacent faces and black on the faces opposite to the red faces and green on the remaining faces is cut into sixty four smaller cubes of equal size.

88. How many cubes are there which have no face painted?
(1) 1 (2) 4
(3) 8 (4) 16

89. How many cubes have only one face pained?
(1) 8 (2) 16
(3) 24 (4) 32

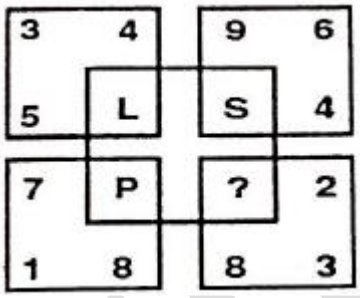
90. How many cubes are there with three faces painted?
(1) 4 (2) 8
(3) 16 (4) 24

91. How many cubes are there with two faces painted?
(1) 24 (2) 8
(3) 32 (4) 12

92. How many cubes have one face green and one of the adjacent faces black or red?
(1) 8 (2) 16
(3) 24 (4) 28

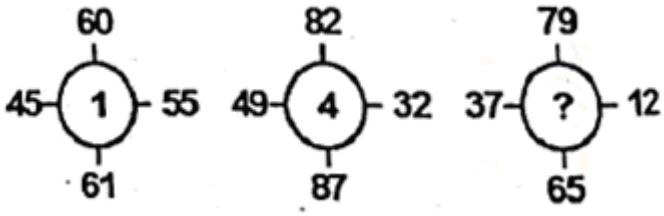
93. Which group of letters is different from others?
(1) CBAED (2) TSRVU
(3) KJIMN (4) WVUYX

94. Find the letter to be placed in place of (?) in the figure given.



(1) M (2) N
(3) Q (4) R

95. Identify the number in the Position of (?)



(1) 4 (2) 5
(3) 6 (4) 7

96. A sprinter goes off the starting block for 100m run and at that instant the second hand of a stop watch had pointed towards North. He touches the finishing line exactly after 12 seconds. In which direction did the second hand point when he just crossed the finishing line?
(1) 18° North of East (2) 18° East of North
(3) 72° North of East (4) 82° East of North

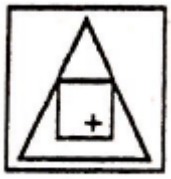
Direction (Questions 97 – 98): A lady runs 12 km towards North, then 6 km towards south and then 8 m East.

97. How far is she from her starting point?
(1) 26 km (2) 18 km
(3) 14 km (4) 10 km

98. Which direction is she from her starting point?
(1) North-East (2) North
(3) East (4) North-West

Direction (Questions 99 – 100): Find out how will the key figure (X) look like after rotation?

99.

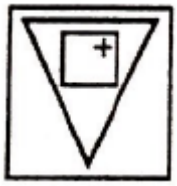


(X)

(1)

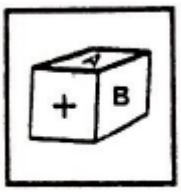
(2)

(3)



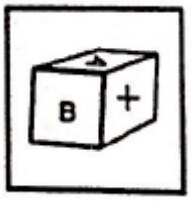
(4)

100.

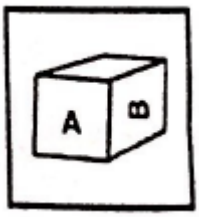


(X)

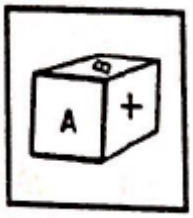
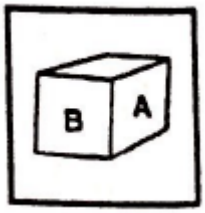
(1)



(2)



(3)



(4)